

Lesson Objective

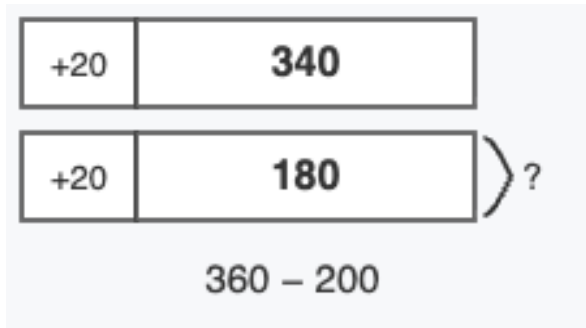
Use the associative property to subtract from three-digit numbers.

Materials: Linking cubes in two colors, personal white board

1. Explore how adding the same amount to both numbers in a subtraction expression keeps the difference the same using linking cubes. Build a stick of 7 cubes and a stick of 4 cubes. Imagine each cube is worth 10. What is the difference? Now add 1 cube to each stick. What is the new subtraction sentence? Did the difference change? Add 1 more cube to each stick. What is the new subtraction sentence? Did the difference change? <i>Answer: $70 - 40 = 30$; $80 - 50 = 30$, the difference did not change; $90 - 60 = 30$, the difference did not change.</i>	Build a stick of 7 cubes in one color and a stick of 4 cubes in a second color. Tell students each cube is worth 10, and count the sticks by tens together. Ask students to name the subtraction sentence for the difference between the two sticks. Add 1 cube of a new color to the end of each stick. Ask students to name the new subtraction sentence and whether the difference changed. Repeat by adding 1 more cube to each stick. Ask students to name the new subtraction sentence and explain why the difference is still 30. Point out that as long as we add the same amount to both numbers, the difference stays the same.
2. Subtract from a three-digit number by adding to both numbers to make a friendly multiple of ten.	Write $340 - 180$ on the board. Ask students how close 180 is to the next friendly hundred and how much they would need to add to 180 to make 200.

Solve $340 - 180$. Use a tape diagram to add the same amount to both numbers to make the problem easier, then write a new subtraction sentence and solve.

Answer: Add 20 to both numbers; $360 - 200 = 160$, so $340 - 180 = 160$.



Guide students to draw a two-bar tape diagram with 340 on top and 180 on the bottom. Have students add 20 to the left side of each bar to show compensation.

Ask students to name the new, easier subtraction sentence and solve it: $360 - 200 = 160$.

Tell students that since they added 20 to both numbers, the difference did not change, so $340 - 180$ also equals 160.

3. Subtract from a three-digit number by adding to both numbers to make a friendly hundred.

Solve $423 - 198$. Add the same amount to both numbers to make a friendly hundred, then write a new subtraction sentence and solve.

Answer: Add 2 to both numbers; $425 - 200 = 225$, so $423 - 198 = 225$.

Write $423 - 198$ on the board. Ask students which friendly hundred 198 is close to and how much they need to add to 198 to make 200.

Have students draw a two-bar tape diagram and add 2 to each bar to show the compensation.

Ask students to write the new subtraction sentence and solve: $425 - 200 = 225$.

Remind students that the difference did not change because the same amount was added to both numbers, so $423 - 198 = 225$.

Monitor For • In Problem 1, can students name each new subtraction sentence ($70 - 40$, $80 - 50$, $90 - 60$) and state that the difference is still 30? • In Problem 2, do students add 20 to both numbers, or do they add 20 only to 180 and forget to also add it to 340? • In Problem 3, do students identify 200 as the friendly hundred close to 198, or do they try to compensate to 100 or another number? • Do students fluently subtract multiples of ten and one hundred from three-digit numbers (e.g., $360 - 200$, $425 - 200$)?	Instructional Moves • In Problem 1, after adding a cube to each stick, point to both sticks and ask students to count by tens on each one so they see that the gap between the tops of the sticks stays the same length. • If a student adds the compensation amount to only one bar in Problem 2, point to both bars and ask, "If we add 20 here, what do we have to do to the other bar to keep the difference the same?" • For Problem 3, prompt students to say how many more 198 needs to reach 200 before they touch the minuend. Use the question stem, "How many more does 198 need to make 200?" • If a student is unsure of the new subtraction sentence, have them write the original problem and the new problem one above the other and circle the matching difference (e.g., $340 - 180 = 160$ and $360 - 200 = 160$).
Reflection Questions • When we added a cube to each stick in Problem 1, what happened to the difference? <i>Answer: The difference stayed the same.</i>	

- In Problem 2, how did you decide how much to add to both numbers? *Answer: I added 20 because 180 needs 20 more to make 200, a friendly multiple of ten.*

- Why does $423 - 198$ give the same answer as $425 - 200$? *Answer: Because I added 2 to both numbers, so the difference didn't change.*

- How does the compensation strategy make subtraction easier? *Answer: It changes the number we are taking away into a friendly ten or hundred, so we can subtract in our head.*